

# Noah Krever

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## RESEARCH INTERESTS

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Algorithmic Game Theory, Computational Economics, Optimization, Uncertain & Online Learning, AI for Society

## EDUCATION

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Columbia University | B.A. in Data Science (CS & Statistics) with Mathematics concentration

May 2022

Major GPA: 3.90/4, Cumulative GPA: 3.78/4

## PUBLICATIONS

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(P: Preprint & Manuscript, C: Conference & Journal, W: Workshop & Demo, \*: Equal contribution, †: Equal senior role)

[C1] [GUARD: Constructing Realistic 2-Player Matrix & Security Games for Benchmarking Game-Theoretic Algorithms](#)

Noah Krever, Jakub Černý, Moïse Blanchard, Christian Kroer

Conference on Neural Information Processing Systems (NeurIPS) 2025

🏆 **Spotlight (Top 3% in Datasets & Benchmarks)**

[P1] [Is Quasar PG1302-102 Periodic?](#)

Noah Krever, Maria Charisi, Daniel D'Orazio, Zoltan Haiman

Manuscript in prep. for *APJL* or similar

## RESEARCH EXPERIENCE

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Columbia IEOR, Lily Xu Group | NY, NY

Researcher, Optimization and AI

Oct. 2025 – Present

- Optimizing monitoring resources for verifiable eco-credit allocation mechanisms and decision protocols
- Modeling metrics & sensor readings for MDPs, multi-armed bandits, and game-theoretic data-auditing

Columbia IEOR, Christian Kroer Group | NY, NY

Researcher, Algorithmic Game Theory

June 2024 – Dec. 2025

- Published first-author spotlight paper introducing statistical theorems and our novel game theory framework
- Implemented original suite of standard algorithms for Nash and Stackelberg equilibria, mixed-integer linear programming, incremental strategy generation (double oracle solvers), and self-play with no-regret learners
- Built flexible object-oriented framework for converting real-world data into hyper-realistic security game instances

Columbia University Physics Department, Zoltan Haiman Group | NY, NY

Researcher, Computational Astrophysics

Jan. 2019 – May 2022

- Lead project studying false periodicities in quasar time-domain surveys, evaluated binary black-hole candidate
- Utilized HPC cluster for large-scale time-series simulations & machine learning toolkits for statistical analysis
- Scripted algorithms to perform spectral analysis, Markov processes, signal processing, and Bayesian model-fitting

TCNJ Physics Department, Tuan Nguyen Lab | Ewing, NJ

Summer Researcher, Computational Biophysics

June 2017 – Aug. 2017

- Mapped network connectivity using single cell laser scanning photo stimulation with calcium imaging
- Wrote original algorithms for automating network analysis, cut processing time from one week to 3 seconds

## WORK EXPERIENCE

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**Roc360** | NY, NY

*Data Scientist*

June 2022 – Present

- Solving open-ended problems in Markov processes, graph theory, network modeling & algorithms
- Training, deploying, automating wide range of machine learning models for classification, ranking, prediction
- Leading development of SQL/Python analytics data layer linking pipelines and models using dbt

**Poker** | NY, NY & Philadelphia, PA

*Semi-Professional Player & Coach*

Sept. 2021 – Present

- Coaching a selective group of ~10 highly motivated students weekly through game-theory optimal (GTO) poker principles, solver study, database analysis, leak-finding, hand reviews
- Playing mid-stakes NLHE cash games semi-professionally, online \$1/\$2 and live up to \$10/\$20
- Creating online educational video-form content for poker strategy improvement

## AWARDS AND HONORS

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**Spotlight Paper (Top 3%), NeurIPS Main Conference**

2025

GUARD: Constructing Realistic Two-Player Matrix and Security Games for Benchmarking Game-Theoretic Algorithms

**Dean's List, Columbia University**

2019 – 2022

Awarded to the Columbia University students maintaining highest GPAs

**Kluge Scholar, Columbia University**

2018 – 2022

Scholarship program offered to select high-achieving students, grants funding for research and internships

## TEACHING EXPERIENCE

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**Artificial Intelligence (COMS 4701)**

Spring 2022

Teaching Assistant at Columbia University (Instructor: Prof. Tony Dear)

**Intro to Computing in Python (ENGI 1006)**

Fall 2020, Fall 2021

Teaching Assistant at Columbia University (Instructor: Prof. Daniel Bauer)

## ACADEMIC SERVICE

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**Program Committee Member**

7th International Workshop on Agents for Societal Impact (ASI, formerly AASG) @ AAMAS

2026

## SELECTED SIDE PROJECTS

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**Regret Minimization in 3-Player Poker** – Customized CFR+ to solve a simplified variant of post-flop 3-player poker and displayed/verified output strategies. Presented for course IEOR 4530: *AI, Games, and Markets*. **Grade:** 100/100

**Understanding Neural ODEs for Time Series** – Implemented novel Neural ODE architectures with adjoint sensitivity training for time series forecasting. Presented for course COMS 4995: *Neural Networks & Deep Learning*. **Grade:** 92/100

**Traversing Latent Spaces for Interpretability** – Novel exploration of deep classifiers' latent spaces, probed decision-boundaries with perturbations & class. loss VAE. Presented for course COMS 6998: *Adv. Deep Learning*. **Grade:** 100/100

## ACTIVITIES

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**Columbia Undergraduate Science Journal (CUSJ), Editor & Board Member for CU Research Journal**

Oct. 2018 – May 2022

**Columbia Poker Club, Founder & President (500+ Member Club Organizer, Teacher, Player)**

July 2021 – May 2022

**Columbia University Esports Club, Rocket League Team Captain & Founder (Top 0.1% of players)**

June 2020 – May 2022

**Columbia University Chess Club, Member (2200 online ELO)**

Sept. 2020 – May 2022